

《数据库系统原理》实验报告（实验三）

题目：实验三：数据库安全性

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实验环境： Docker OceanBase

实验步骤及结果截图：

0. 建立数据库 Major 并使用

```
obclient(root@sys)[(none)]> create database major;
Query OK, 1 row affected (0.041 sec)
```

```
obclient(root@sys)[(none)]> use major;
Database changed
```

1.在 Major 数据库中创建表 Players

```
obclient(root@sys)[major]> create table players (
->   sno varchar(10) primary key,
->   sname varchar(20) unique,
->   ssex varchar(1),
->   sage smallint,
->   steam varchar(20)
-> );
Query OK, 0 rows affected (0.085 sec)
```

```
obclient(root@sys)[major]> desc players;
+-----+-----+-----+-----+-----+-----+
| Field | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| sno   | varchar(10)   | NO   | PRI | NULL    |       |
| sname | varchar(20)   | YES  | UNI | NULL    |       |
| ssex  | varchar(1)    | YES  |     | NULL    |       |
| sage  | smallint(6)   | YES  |     | NULL    |       |
| steam | varchar(20)   | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.007 sec)
```

2.向 Players 表中插入两个元组：

('200215121','LuLu','M',27,'G2')

('200215122','TiMo','M',18,'G2')

```
obclient(root@sys)[major]> insert into players values
-> ('200215121', 'LuLu', 'M', 27, 'G2'),
-> ('200215122', 'TiMo', 'M', 18, 'G2');
Query OK, 2 rows affected (0.010 sec)
Records: 2 Duplicates: 0 Warnings: 0
```

```
obclient(root@sys)[major]> select * from players;
+-----+-----+-----+-----+-----+
| sno   | sname | ssex | sage | steam |
+-----+-----+-----+-----+-----+
| 200215121 | LuLu | M   | 27  | G2    |
| 200215122 | TiMo | M   | 18  | G2    |
+-----+-----+-----+-----+-----+
2 rows in set (0.002 sec)
```

3.建立用户 Jay，密码设置为学号 2351520，授予用户 Jay create table 的权限，并赋予其再授权的能力

```
obclient(root@sys)[(none)]> create user jay identified by '2351520';
Query OK, 0 rows affected (0.028 sec)
```

```
obclient(root@sys)[(none)]> grant create
-> on major.players
-> to jay
-> with grant option;
Query OK, 0 rows affected (0.022 sec)
```

4.在 Jay 用户下尝试查询 Players（注意使用 Major.Players）

```
sh-4.4# obclient -h 127.1 -p2881 -u jay -p
Enter password:
Welcome to the OceanBase. Commands end with ; or \g.
Your OceanBase connection id is 3221487634
Server version: OceanBase_CE 4.2.1.10 (r110000072024112010-28c1343085627e79a4f13c29121646bb889cf901)
```

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

```
obclient(jay@(none))[(none)]> select * from major.players;
ERROR 1142 (42000): SELECT command denied to user 'jay'@'%' for table 'players'
[172.17.0.2:2882] [2025-04-22 08:28:25.286712] [YB42AC110002-000633592B3EA625-0-0]
```

5.授予用户 Jay 对表 Players 的查询、插入、修改等对象特权，并赋予其再授权的能力

```
obclient(root@sys)[(none)]> grant select,insert,update
-> on major.players
-> to jay
-> with grant option;
Query OK, 0 rows affected (0.020 sec)
```

6.在 Jay 用户下查询 Players（注意使用 Major.Players）

```
obclient(jay@(none))[(none)]> select * from major.players;
+-----+-----+-----+-----+-----+
| sno      | sname | ssex | sage | steam |
+-----+-----+-----+-----+
| 200215121 | LuLu  | M    | 27   | G2     |
| 200215122 | TiMo  | M    | 18   | G2     |
+-----+-----+-----+-----+
2 rows in set (0.003 sec)
```

7.在 Jay 用户下再插入一个元组的数据('200215123','ZywOo','M',23,'Vit')，并在 Jay 用户下查询（注意使用 Major.Players）

```
obclient(jay@(none))[(none)]> insert into major.players values('200215123','ZywOo','M',23,'Vit');
Query OK, 1 row affected (0.009 sec)
```

```
obclient(jay@(none))[(none)]> select * from major.players;
+-----+-----+-----+-----+-----+
| sno      | sname | ssex | sage | steam |
+-----+-----+-----+-----+
| 200215121 | LuLu  | M    | 27   | G2     |
| 200215122 | TiMo  | M    | 18   | G2     |
| 200215123 | ZywOo | M    | 23   | Vit    |
+-----+-----+-----+-----+
3 rows in set (0.000 sec)
```

8.删除 Jay 用户

```
obclient(root@sys)[(none)]> drop user jay;
Query OK, 0 rows affected (0.021 sec)
```

出现的问题:

为了修改用户密码, 创建用户 **jay** 时报错, 提示无法创建用户

```
obclient(root@sys)[(none)]> create user jay identified by '2351520';  
ERROR 1396 (HY000): Operation CREATE USER failed for 'jay'@'%'  
[172.17.0.2:2882] [2025-04-22 08:52:11.834088] [YB42AC110002-000633592B3EA63D-0-0]  
obclient(root@sys)[(none)]>
```

解决方案:

发现是因为之前已经创建过用户 **jay**。需要将原来的用户删除再重新创建

```
obclient(root@sys)[(none)]> drop user jay;  
Query OK, 0 rows affected (0.025 sec)  
  
obclient(root@sys)[(none)]> create user jay identified by '2351520';  
Query OK, 0 rows affected (0.025 sec)
```